Business Understanding Report — Customer Churn Analysis

1. Business Context

Customer churn occurs when customers discontinue their relationship with a company. High customer churn can negatively affect revenue, customer lifetime value and business growth because the organization must continually replace customers who leave.

For a telecommunications business, understanding why customers leave is particularly important because customers can choose between different service providers, contract types, payment methods and additional services.

This project analyses customer churn data to identify patterns and customer segments associated with higher churn. The objective is to provide management with data-driven insights that can support customer retention strategies.

2. Business Problem

The organization is experiencing customer churn and needs to understand which customer characteristics and services are associated with customers leaving.

The analysis therefore focuses on seven key business questions:

1. What does the customer base look like?
2. Which customer segments have the highest churn?
3. Does contract type influence retention?
4. Does tenure affect customer loyalty?
5. Which services are associated with churn?
6. Which payment methods have higher churn?
7. What actions should management take?

3. Analytical Objectives

The analysis aims to:

- Understand the composition of the customer base.
- Identify high-risk customer segments.
- Examine the relationship between contract type and churn.
- Analyze how customer tenure relates to churn.

- Assess differences in churn across services.
- Compare churn across payment methods.
- Translate findings into practical management recommendations.

4. Data and Tools

The analysis was conducted using **Microsoft Excel**, with the dataset cleaned and structured into an Excel table. Descriptive analysis and visualizations were created to examine customer characteristics and churn patterns.

The analysis includes bar charts, pie charts, histograms, a box-and-whisker plot and a correlation heatmap.

5. Key Business Findings

The analysis indicates that churn is not evenly distributed across the customer base.

Month-to-month customers have a **42.71% churn rate**, compared with **11.27%** for one-year contracts and **2.83%** for two-year contracts.

Tenure also shows a strong relationship with churn. Customers with 0–12 months of tenure have the highest churn rate at **47.44%**, while customers with 61–72 months of tenure have a churn rate of only **6.61%**.

Fiber optic customers have a **41.89% churn rate**, considerably higher than DSL customers at **18.96%**.

Customers without additional services also tend to have higher churn. For example, Online Security customers without the service have a churn rate of **41.77%**, compared with **14.61%** among customers with the service.

Payment method also shows a notable difference. Customers using electronic checks have the highest churn rate at **45.29%**, compared with **15.24%** among customers using automatic credit-card payments.

6. Business Implications

The findings suggest that management should prioritize **early-stage retention**, particularly during the first 12 months of the customer relationship.

The month-to-month segment also represents an important retention opportunity because it combines a large customer base with high churn.

High churn among fiber optic and electronic-check customers should be investigated further to determine the underlying causes rather than assuming that these characteristics directly cause churn.

7. Conclusion

The analysis demonstrates that customer churn is concentrated within identifiable customer segments. Management can use these findings to develop targeted retention strategies rather than applying a single approach to the entire customer base.

The recommended focus areas are early customer engagement, contract conversion, investigation of high-risk service segments, payment experience improvements and targeted value-added service offers.